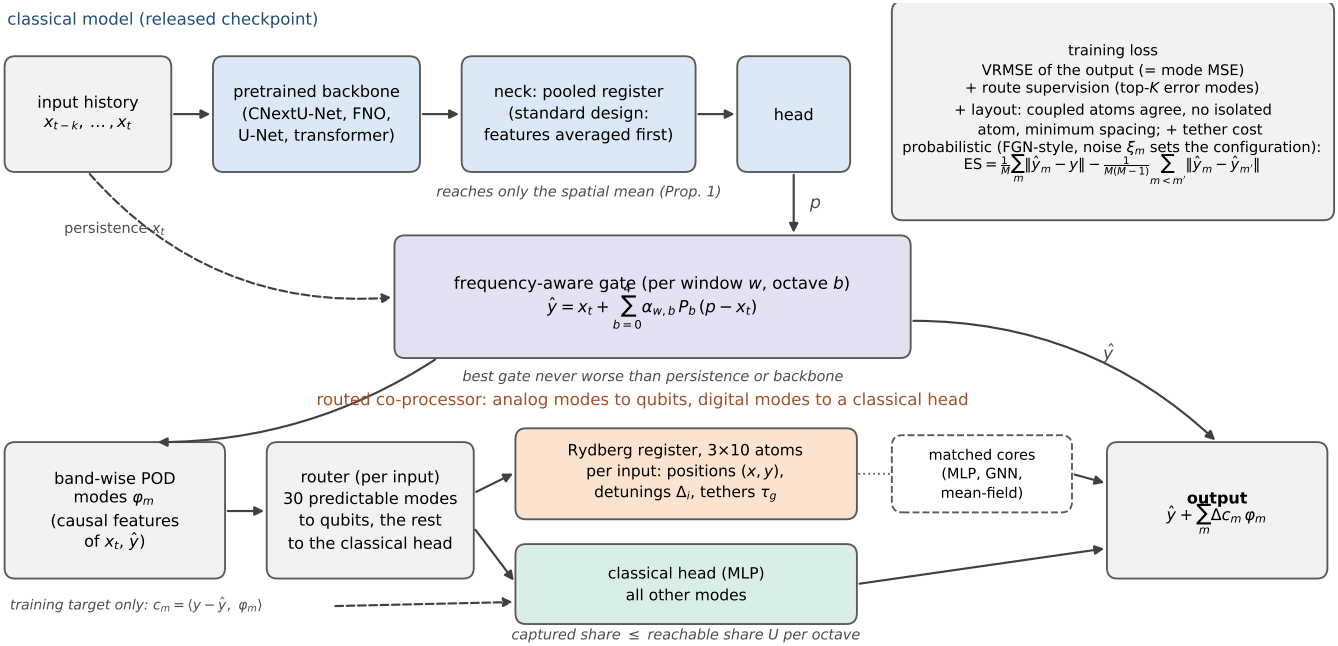
classical model (released checkpoint)
input history
$x_{t-k}, \ldots, x_t$
pretrained backbone
(CNextU-Net, FNO,
U-Net, transformer)
neck: pooled register
(standard design:
features averaged first)
head
reaches only the spatial mean (Prop. 1)
p
persistence x_t
training loss
VRMSE of the output (= mode MSE)
+ route supervision (top-K error modes)
+ layout: coupled atoms agree, no isolated
atom, minimum spacing; + tether cost
probabilistic (FGN-style, noise ξ_m sets the configuration):
$\mathrm{ES} = \frac{1}{M}\sum_m \|\hat{y}_m - y\| - \frac{1}{M(M-1)}\sum_{m<m'} \|\hat{y}_m - \hat{y}_{m'}\|$
frequency-aware gate (per window w, octave b)
$\hat{y} = x_t + \sum_{b=0}^{L} \alpha_{w,b} P_b (p - x_t)$
$\hat{y}$
best gate never worse than persistence or backbone
routed co-processor: analog modes to qubits, digital modes to a classical head
band-wise POD
modes φ_m
(causal features
of x_t, $\hat{y}$)
router (per input)
30 predictable modes
to qubits, the rest
to the classical head
Rydberg register, 3×10 atoms
per input: positions (x, y),
detunings Δ_i, tethers τ_g
matched cores
(MLP, GNN,
mean-field)
classical head (MLP)
all other modes
output
$\hat{y} + \sum_m \Delta c_m \varphi_m$
training target only: $c_m = \langle y - \hat{y},\ \varphi_m \rangle$
captured share $\leq$ reachable share U per octave